

Emmanuel Tega Agbragu

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SUMMARY

Security engineer with 4+ years across offensive and defensive security, combining manual web and API penetration testing with enterprise security infrastructure deployment for regulated banking institutions. Proven track record of identifying and validating vulnerabilities as well as producing formal risk assessments aligned to industry standards and frameworks

EXPERIENCE

Security Researcher

Jan 2025 – Apr 2026

The Upanzi Network, Carnegie Mellon University | Kigali, Rwanda

- Conducted penetration tests of web applications and APIs across e-government and research platforms, identifying and exploiting critical access-control, business-logic, and injection vulnerabilities with hands-on proof-of-concept verification.
- Discovered and exploited a mass-assignment flaw enabling privilege escalation to administrator and a BOLA/IDOR vulnerability rated **Critical (CVSS 9.1)**, permitting full unauthorized creation, retrieval, modification, and deletion of product data on an e-commerce API.
- Identified an indirect prompt-injection vulnerability in an LLM-powered dataset feature, demonstrating model behavior manipulation via crafted PDF uploads.
- Authored formal VAPT reports per engagement, documenting risk ratings aligned to OWASP and NIST, tracking remediation through to verified closure with development teams.
- Built custom Python automation scripts and extended APKit (open-source mobile vulnerability scanner) to scale endpoint enumeration and vulnerability detection across large asset inventories.

Graduate Teaching Assistant

Sep 2024 – Jan 2025

Carnegie Mellon University Africa | Kigali, Rwanda

- Delivered lab sessions in Ethical Hacking and Cybersecurity Operations, reinforcing student competency in offensive and defensive security.

Cybersecurity Engineer

Apr 2021 – Jun 2023

Lumenave International Limited | Lagos, Nigeria

- Designed and deployed security infrastructure (IDS/IPS, EDR, firewalls) for banking clients, reducing malware infections by **40%**.
- Implemented CyberArk PAM solutions with role-based access controls and least privilege policies for compliance requirements.
- Configured Symantec DLP policies defining detection rules and access controls to prevent unauthorized modification, disclosure, and exfiltration of sensitive financial data.
- Led a team of 5 engineers delivering security projects on time and under budget constraints.
- Oversaw smooth operations for all deployed security solutions by providing on-site and off-site technical support, resulting in **100%** of the operational efficiency of the solutions.

SELECTED PROJECTS

Securing the Digital State — Research Report

2025

Python, Automation | Upanzi Network

- Contributed to a pan-African study assessing the security posture of **21,782** government sub-domains across 54 countries.
- Led HTTP-methods enumeration, identifying dangerous exposed methods.
- Built custom automation scripts powering mass asset discovery and scanning.

Privileged Access Management Deployment

2021 – 2023

CyberArk, Bash, Oracle, PowerShell | First City Monument Bank

- Deployed a complete CyberArk stack managing privileged account access across network perimeters, with secure authentication and least-privilege role-based access controls aligned to regulatory requirements.

TECHNICAL SKILLS

Penetration testing	Web & API penetration testing, business-logic flaws, access-control
Security Tools	Burp Suite, OWASP ZAP, Nmap, Nessus, OpenVAS, SQLMap
Web & Networking	HTTP, TCP/IP, REST APIs, authentication (SAML, OAuth, LDAP, Kerberos)
Programming	Python, Bash, C/C++, Java, SQL, PHP
Frameworks	OWASP, MITRE ATT&CK, NIST, ISO 27001, PCI-DSS

EDUCATION

MSc Information Technology

Aug 2023 – May 2025

Carnegie Mellon University | GPA: 3.80/4.0 | Mastercard Foundation Partial Scholarship

BSc Pure and Applied Physics — First Class Honours

Mar 2014 – May 2018

University of Benin | Nigeria | GPA: 4.54/5.0 | Top 1% of graduating class

CERTIFICATIONS AND ACHIEVEMENTS

CEH Master • eJPT • AWS Certified Cloud Practitioner • OSCP (In Progress) • Top 10 Finalist, Greenfist CTF 2023 • Mastercard Foundation Scholarship 2023 • First Class Honours, Top 1%, 2018